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CSC 21000 HW 4

OBJECTIVE: Our Objective was to Create a rational number class and create a fixed point class

RESULTS:

```
/*  
Hello students, the homework for the upcoming week is the following: 1- Create a  
rational number class and 2- Create a fixed point class. Make sure to implement  
all the methods for all the operations that we can apply to those numbers.  
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*/  
#include <iostream>  
#include <bitset>  
#include <cmath>  
  
using namespace std;  
class FixedPoint  
{  
private:  
int size;  
int point;  
int diff;  
int num;  
double Result;  
  
public:  
FixedPoint(int siz, int poi, int number){  
size=siz;  
point=poi;  
num=number;  
};  
double GetBinaryResult(){return Result;}  
  
void GetFixedpoint()  
{  
diff=size-point;  
string binarystring="";  
for(int i=size-1;i>=0;i--){  
binarystring+= to_string(num % 2 );  
num= num/2;  
}  
}
```

```

    string revBinarystring=string(binarystring.rbegin(),binarystring.rend());
    string BinaryInt= string(revBinarystring.begin(), revBinarystring.begin()+diff);
    string BinaryPoint=string(revBinarystring.begin()+diff,revBinarystring.end());

    string revBinaryInt=string(BinaryInt.rbegin(),BinaryInt.rend());
    int intresult;//loop for binaryint
    for(int i=0;i<BinaryInt.size();i++){
        intresult+= stoi( revBinaryInt.substr(i,1) )*pow(2,i);
    }

    double doubleresult;//loop for binaryint
    for(int i=0;i<BinaryPoint.size();i++){
        doubleresult+= stod( BinaryPoint.substr(i,1) )*pow(2,(-1*i)-1);
    }
    Result=intresult+doubleresult;

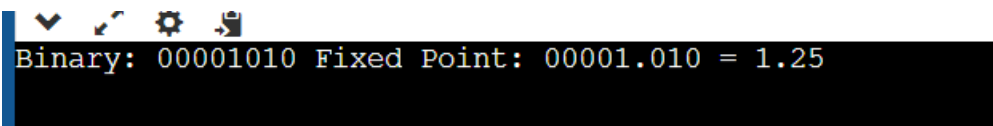
    cout<<"Binary: "<<revBinarystring<<" Fixed Point: "<< BinaryInt<< "."<<BinaryPoint<<" = "<<Result<<endl;
}

};

int main()
{
    int number=10;
    int size=8;
    int point=3;
    FixedPoint binary1(size,point,number);
    binary1.GetFixedpoint();

    return 0;
}

```



```

Binary: 00001010 Fixed Point: 00001.010 = 1.25

```

EXPLANATION: Unfortunately I was confused on what to do for the rational number class so I did not do the problem.